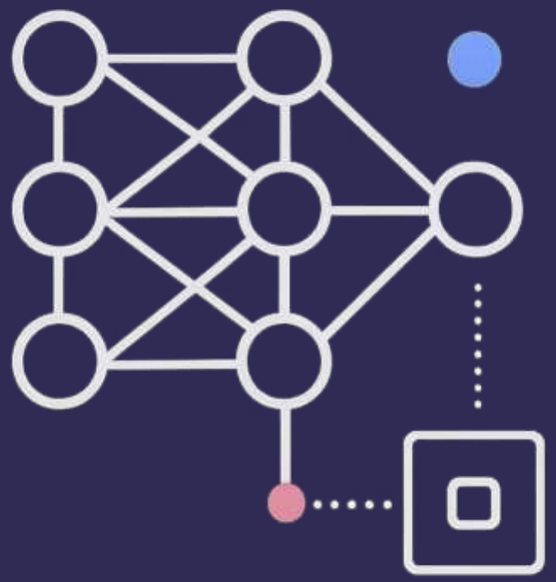

Fake News Detection System

Parth Azad | Data Engineering & AI/ML Project

PySpark • LSTM • FAISS • Gemini

Made with GAMMA

The Challenge: Fighting Information Warfare

Exponential Growth

Fake news spreads 10x faster than legitimate information on social platforms

Verification Crisis

Manual fact-checking cannot scale with volume of content generated daily

Explainability Gap

Traditional ML models operate as black boxes without transparent reasoning

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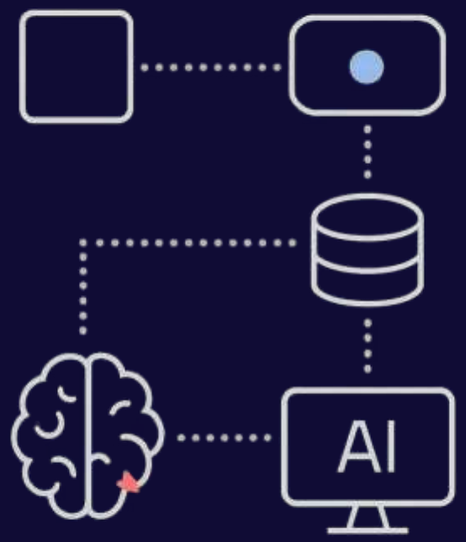
Our Solution: AI-Powered Verification

End-to-End Pipeline

An integrated system combining machine learning, data engineering, and generative AI to deliver accurate, explainable predictions with validation.

Key Capabilities

- Binary classification (Fake/Real)
- Confidence scoring
- Natural language explanations
- Source validation



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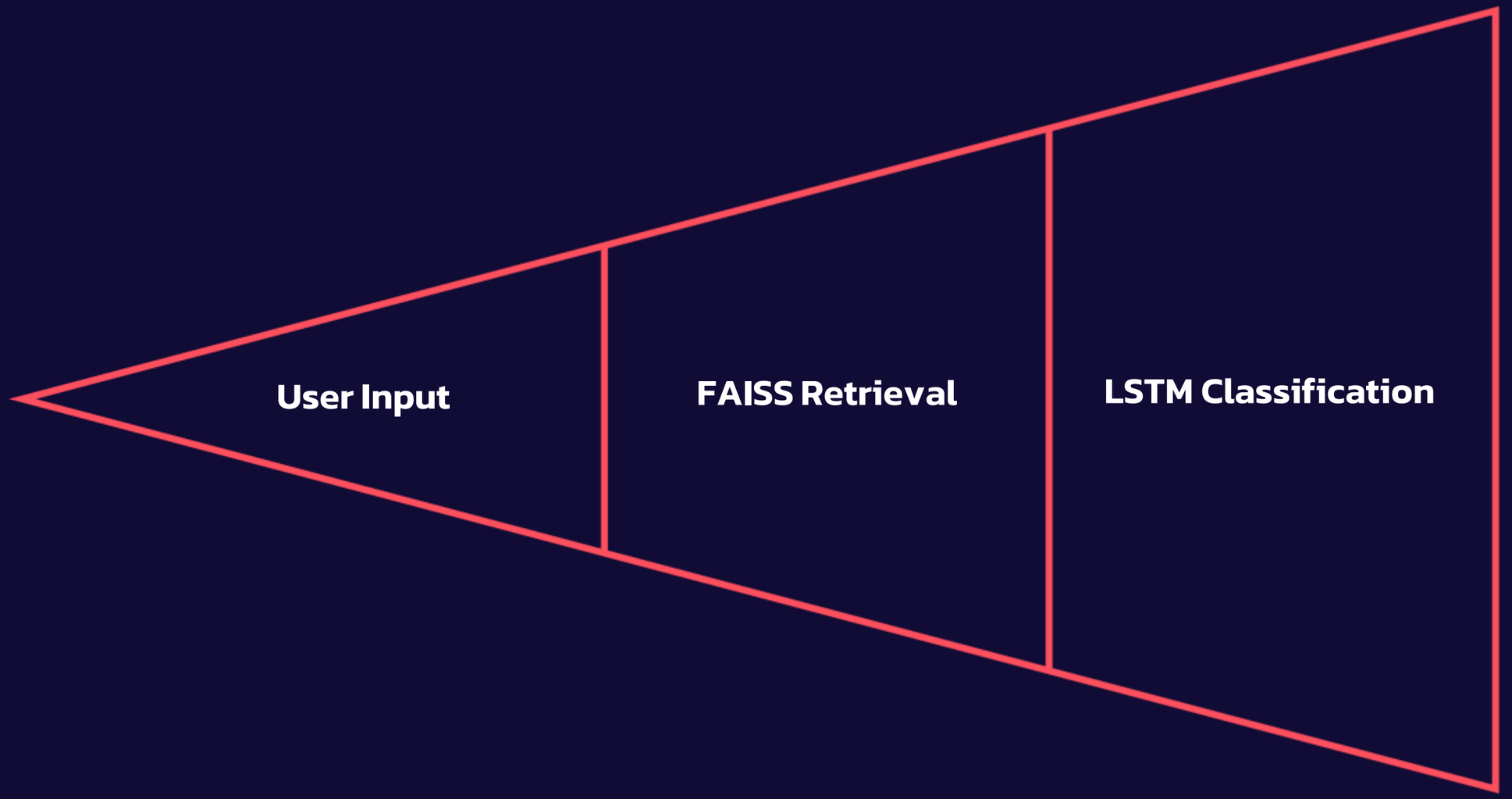
System Architecture



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Data Engineering: Medallion Architecture

Bronze Layer

Raw data ingestion from multiple sources using PySpark

- JSON/CSV collection
- Initial storage in Parquet format

Silver Layer

Data cleaning and preprocessing

- Null value handling
- Regex pattern cleaning
- Text normalization

Gold Layer

Feature engineering for ML

- Lemmatization
- Vector embeddings
- Business Logic

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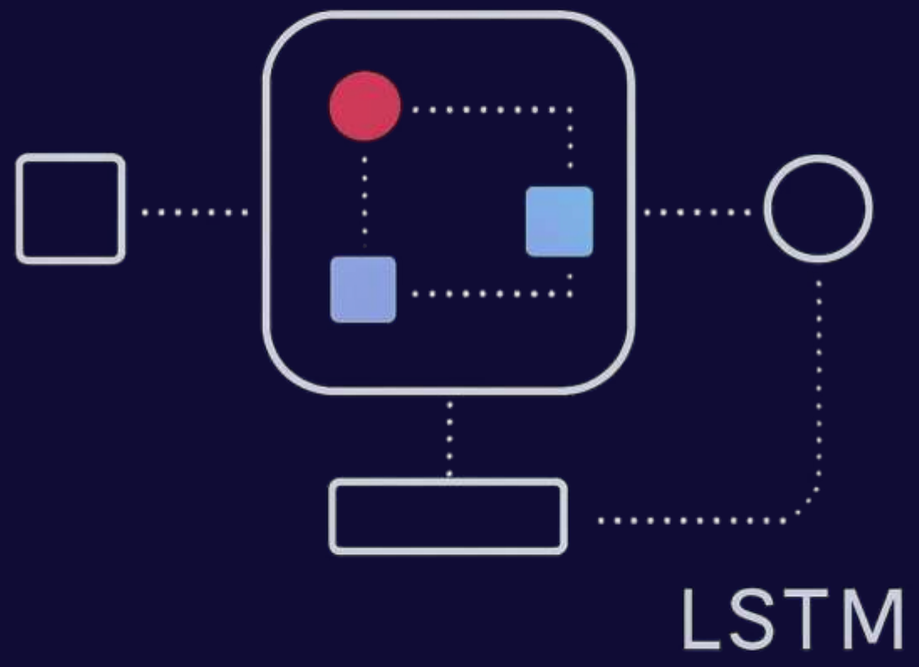
Machine Learning Model: LSTM Architecture

Why LSTM?

Long Short-Term Memory networks excel at capturing sequential context and dependencies in text, making them ideal for understanding narrative patterns that distinguish fake from real news.

Model Pipeline

1. Text tokenization
2. Sequence padding
3. GloVe embedding lookup
4. BiLSTM processing
5. Output probability



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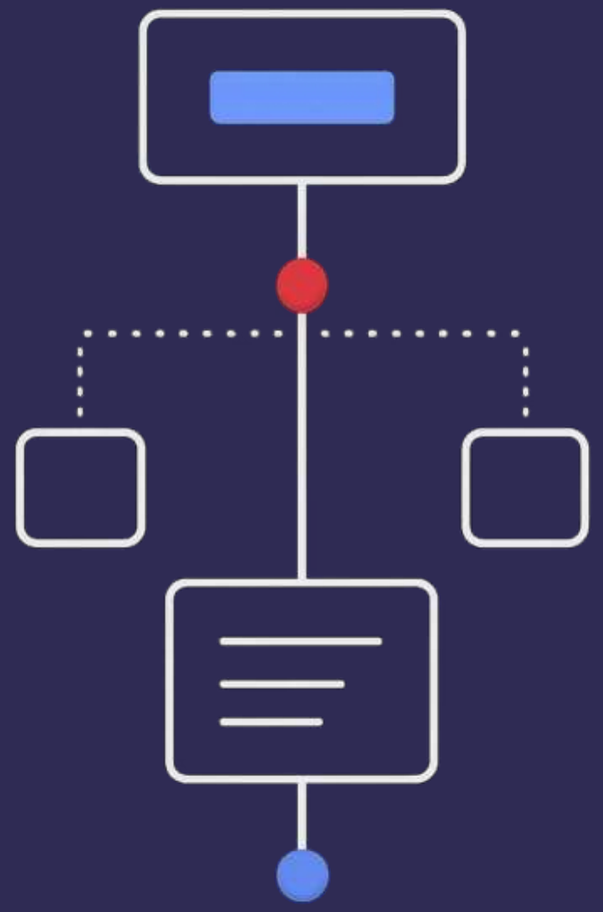

RAG Pipeline: Context-Aware Retrieval

Text Encoding

Convert articles to vector embeddings using transformer models

FAISS Indexing

Store vectors in optimized similarity search structure

Top-K Retrieval

Find most similar articles for context

Validation

Compare predictions with retrieved sources

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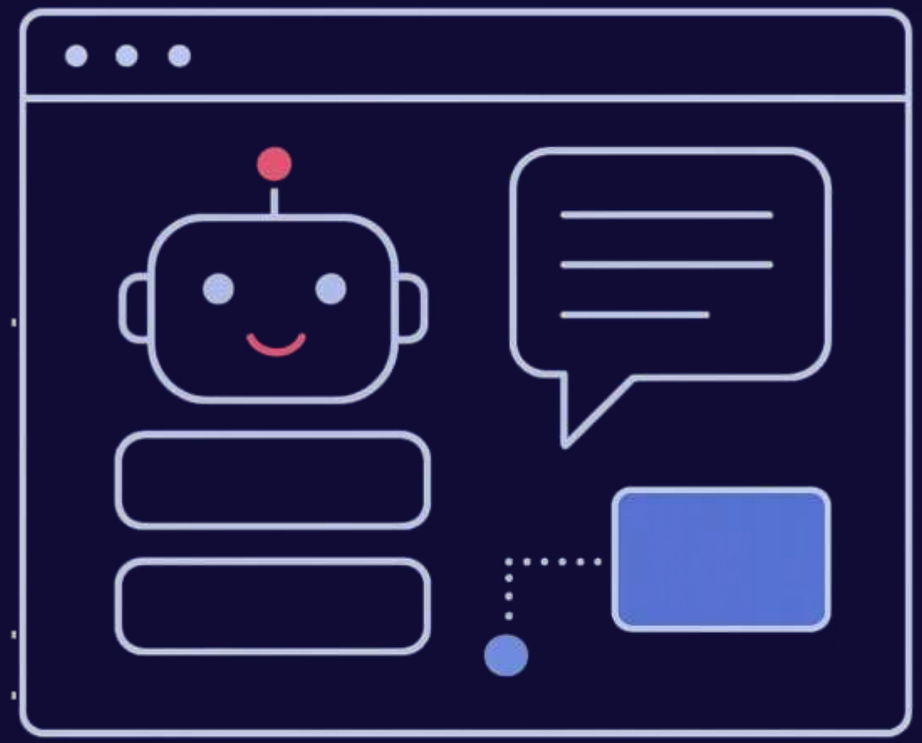
Gemini Integration: Explainable AI



Prompt Engineering

Custom-tuned prompts guide Gemini to analyze LSTM predictions alongside retrieved context, generating comprehensive explanations.

Output Generation

- Plain-English reasoning
- Source consistency analysis
- Confidence justification
- Error flagging when uncertain

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Results & Performance

92%

Model Accuracy

Achieved 90-94% accuracy across multiple train-test splits

15%

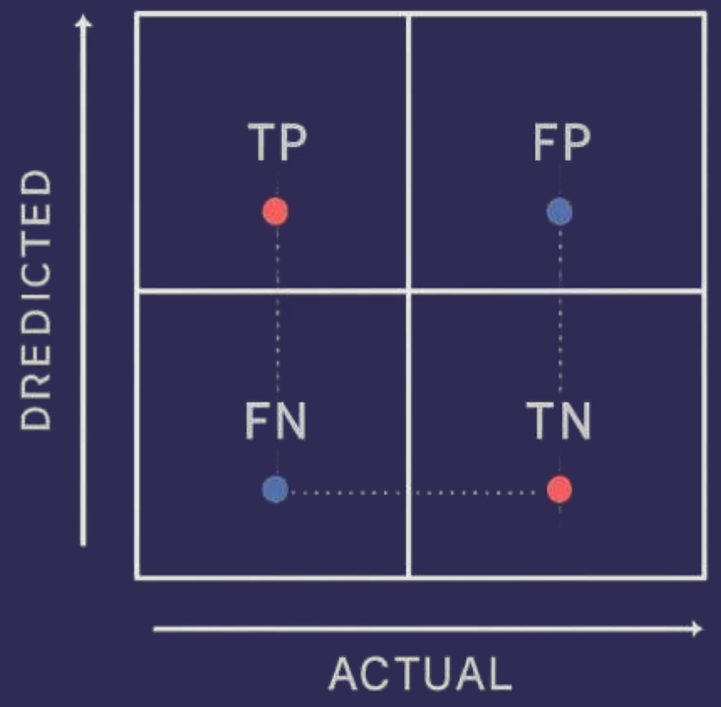
False Positive Reduction

RAG integration reduced false positives vs. standalone LSTM

87%

Average Confidence

High confidence scores for validated predictions



Example Output

> **Prediction:** Fake | **Confidence:** 0.87 | **Explanation:** Inconsistencies detected with 87% of retrieved sources citing contradictory facts about timeline and participants

Future Roadmap

1. **Production Deployment**

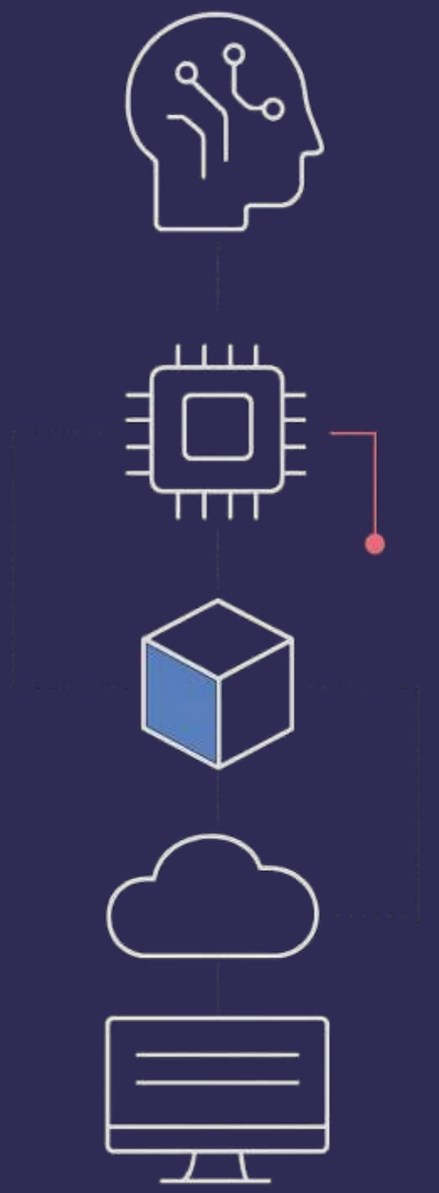
   Deploy on Databricks for enterprise-scale processing

2. **Real-Time API**

   Build REST interface for live article verification

3. **BERT Integration**

   Replace LSTM with transformer-based models for better performance

4. **Advanced RAG**

   Upgrade to Pinecone or Weaviate for better vector search